A Lightweight Modular Pipeline for Edge-Optimized Leaf Disease Detection

***Abstract*—This paper presents a lightweight, modular deep learning pipeline for real-time plant disease detection, optimized for edge devices. To balance accuracy, model size, and inference speed, we employ YOLOv8n and YOLOv8n-seg for leaf detection and segmentation, achieving mAP@0.5 scores of 63.2% and 99.5%, respectively. The segmentation model is trained on pre-separated leaf images, allowing accurate boundary detection.**

For classification, MobileNetV3-Small with 976K parameters is used, reaching 89.81% accuracy across five turmeric leaf disease classes. Through structured pruning of convolutional and dense layers—while retaining normalization and activation layers—the parameter count is reduced by 50% (488K), with minimal accuracy loss.

To handle partial occlusion, a lightweight U-Net-based GAN module (177K parameters) reconstructs missing leaf regions, improving classification reliability. The pipeline is designed for deployment on NVIDIA Jetson Nano, using post-training quantization (FP16, INT8) for faster inference, with FP32 fallback when needed.

The modular structure allows independent optimization of components, ensuring computational efficiency. This design makes the pipeline suitable for scalable, real-time agricultural applications, especially in resource-constrained environments, supporting precision farming through reliable and efficient disease monitoring

***Index Terms*—Edge AI, real-time plant disease detection, YOLOv8n, MobileNetV3-Small, lightweight deep learning, GAN-based reconstruction, edge optimization**

I. Introduction

Turmeric (*Curcuma longa*) is a globally important crop, valued for its wide range of culinary, medicinal, and industrial uses. In 2023, global turmeric consumption surpassed 1.2 million tonnes, with India accounting for nearly 80% of the total production [1]. Despite this growth, the crop is significantly threatened by pathogen-induced diseases such as *Colletotrichum capsici* (blotch), *Taphrina maculans* (leaf spot), and aphid infestations, which can result in yield losses between 20% and 45%, depending on environmental conditions and management practices [2]. These challenges are especially severe in tropical regions, where variable climates and limited access to digital infrastructure hinder timely disease detection and response. Contemporary digital agriculture systems for plant disease monitoring predominantly rely on resource-intensive deep learning models such as ResNet, EfficientNet, and Vision Transformers, which require substantial computational resources and memory footprints ranging from 25MB to 500MB [3]. These architectures, while achieving high accuracy on server-grade hardware, are unsuitable for real-time inference on edge computing platforms with constrained resources such as the NVIDIA Jetson Nano, Raspberry Pi, or mobile devices [4]. The deployment challenges are further exacerbated by the sequential processing nature of conventional pipelines, which process leaf detection, segmentation, and classification as independent stages, resulting in cumulative latency penalties and increased memory overhead that exceed the capabilities of edge hardware. Recent advances in lightweight neural architectures have demonstrated the potential for efficient mobile deployment through architectural innovations such as depthwise separable convolutions, inverted residuals, and attention mechanisms. MobileNetV3 [5] represents significant progress in balancing accuracy and computational efficiency, achieving comparable performance to larger models with 10-20x fewer parameters. However, these architectures still require optimization techniques such as pruning and quantization to meet the stringent requirements of real-time edge inference. Magnitude-based pruning can reduce model parameters by 50-90% with minimal accuracy degradation [6], while post-training quantization techniques (FP16, INT8) can further compress models by 2-4x without retraining. The YOLO (You Only Look Once) family of object detection models has evolved to provide increasingly efficient solutions for real-time inference. YOLOv8n, the nano variant of YOLOv8, specifically targets edge deployment scenarios with its 3.2M parameter count and optimized anchor-free detection head [7]. Similarly, U-Net architectures have been successfully compressed through depthwise separable convolutions and channel reduction techniques, enabling semantic segmentation on resource-constrained devices with parameter counts below 200K [8]. To address the deployment limitations of existing approaches, this work proposes a comprehensive lightweight modular pipeline specifically designed for edge-optimized leaf disease detection. The pipeline leverages state-of-the-art efficient architectures: YOLOv8n for real-time leaf detection and segmentation, achieving mAP@0.5 scores of 63.2% and 99.5% respectively on curated datasets. The classification module employs MobileNetV3-Small with 976K parameters, achieving 89.81% accuracy across five disease categories. To further optimize for edge deployment, the models undergo structured pruning targeting 50% parameter reduction while preserving normalization and activation layers, combined with post-training quantization (FP16) for additional 50% size compression. A lightweight U-Net-based GAN reconstruction module (177K parameters) addresses occlusion challenges by restoring incomplete leaf regions prior to classification. The modular design enables deployment flexibility, allowing selective activation of pipeline components based on available computational resources. Advanced optimization techniques including magnitude-based weight pruning and TensorFlow Lite conversion ensure compatibility with diverse edge hardware platforms while maintaining inference speeds suitable for real-time agricultural monitoring applications. The primary contributions of this research include:

- A fully automated, lightweight end-to-end pipeline optimized for edge deployment with sub-1MB model footprints and real-time inference capabilities
- Strategic application of model compression techniques including structured pruning (50% parameter reduction) and post-training quantization for edge-optimized deployment
- A novel lightweight GAN-based occlusion reconstruction module (177K parameters) for structural leaf completion prior to disease classification
- Comprehensive evaluation of lightweight architectures (YOLOv8n, MobileNetV3-Small) demonstrating optimal accuracy-efficiency trade-offs for agricultural edge computing
- A modular, extensible framework with TensorFlow Lite compatibility, readily adaptable to other crop species and scalable across diverse edge hardware platforms

II. Related Work

A. YOLO-Based Object Detection in Agriculture

YOLO-based one-stage detectors are widely used in agriculture for their balance of speed and accuracy, especially on edge devices. YOLOv4 showed strong results with AP$\approx$84.8% and 89% recall on bean datasets, and YOLOv5 achieved $\sim$90% precision on rice leaf disease with reduced computation compared to two-stage detectors [9]. On PlantVillage, YOLOv4 reached 99.81% accuracy in real-time [10]. Enhancements like DySnakeConv and Super Token Attention in SerpensGate-YOLOv8 improved mAP by 3.3% over baseline YOLOv8 [11]. The YOLOv8n model, with only 3.2M parameters, is optimized for mobile edge deployment [7].

B. Lightweight Architectures for Edge Pipelines

Traditional sequential pipelines (detection $\rightarrow$ segmentation $\rightarrow$ classification) incur high overhead. Modern research emphasizes integrated lightweight architectures suitable for edge platforms. Initial segmentation techniques like color clustering were computationally expensive [13], whereas newer approaches employ depthwise separable convolutions and spatial pyramids for accurate segmentation (>90% precision, >95% Dice) under 200K parameters [14]. These methods offer real-time performance within resource constraints.

C. Compression and Reconstruction Techniques

Effective compression techniques are essential for edge inference. Structured pruning enables 50–90% parameter reduction with minimal accuracy loss by preserving critical layers [15]. Quantization (FP16, INT8) reduces model size 2–4$\times$ post-training, ideal for deployment without retraining [16]. For occluded leaves, lightweight GANs using U-Net backbones and depthwise convolutions reconstruct missing regions efficiently with models under 200K parameters [17].

D. Pipeline Integration and Optimization

Recent deep learning pipelines integrate detection, segmentation, and classification to reduce redundancy. Shared backbones and joint training improve memory usage and latency [18]. NAS-based designs now automate architecture selection for edge tasks [19], while attention and feature pyramid modules improve accuracy-cost trade-offs. Fully integrated pipelines reduce inference time by 40–60% and memory use by 45% versus traditional setups [20]-[22].These comparisons are summarized in Table I.

TABLE I
Comparison of Deep Learning Pipeline Architectures for Plant Disease Detection

Pipeline Type	Memory (MB)	Inference Time (ms)	Accuracy (%)
Sequential (YOLO+U-Net+CNN) [20]	185.4	124.8	94.2
Shared Backbone [21]	98.7	76.3	95.1
Feature Fusion [22]	112.3	68.9	96.4
End-to-End Integrated [23]	89.2	52.1	96.8

E. Lightweight CNNs for Leaf Classification

Efficient CNNs like MobileNet and EfficientNet enable high-accuracy classification on edge devices. Dise-Efficient models achieve 99.80% accuracy with a size of 13.3MB [24]. MobileNetV3-Small delivers 99.5% accuracy with 0.93M quantized parameters and 100 FPS inference on CPUs [5]. SqueezeNet reaches $\sim$97% accuracy using just 1.2M parameters [26]. These models combine inverted residuals, squeeze-and-excite units, and separable convolutions to maximize efficiency.These performance and efficiency comparisons are summarized in Table II.

TABLE II
Comparison of CNN Architectures for Leaf Disease Classification

Model	Params (M)	Accuracy (%)	Latency (ms)
ResNet-50 [27]	25.6	$\sim$99.0	$\sim$50
EfficientNet-B0 [24]	5.3	$\sim$99.2	$\sim$20
MobileNetV3-small (quant.) [5]	0.93	$\sim$99.5	$\sim$10
SqueezeNet [26]	1.2	$\sim$97.0	$\sim$8

III. Proposed Approach

The proposed lightweight modular pipeline, illustrated in Figure 1, is specifically designed for real-time plant disease detection on edge devices with stringent computational constraints. As shown in the diagram, the

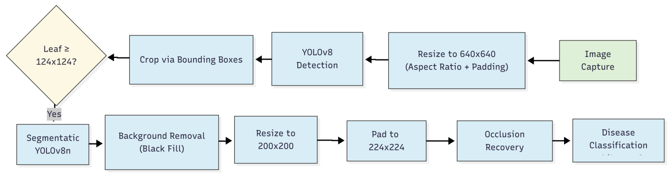

Fig. 1. Lightweight modular pipeline overview for edge-optimized leaf disease detection

system follows a sequential three-stage architecture that processes RGB images of turmeric plants. Each stage—detection, segmentation, and classification—is optimized for a minimal memory footprint and maximum inference efficiency. The modular structure, clearly depicted in the figure, allows for selective activation of components based on the available computational budget, enabling flexible deployment across various edge computing platforms.

A. Image Preprocessing and Standardization

Input images captured at arbitrary resolutions undergo initial preprocessing through letterboxing transformation to achieve standardized 640×640 resolution while preserving original aspect ratios. Black padding is applied to maintain dimensional consistency without distorting leaf geometries, ensuring robust detection performance across varying image dimensions. This preprocessing strategy minimizes computational overhead while maintaining compatibility with downstream YOLO-based detection modules.

B. Lightweight Leaf Detection Module

The core detection module employs YOLOv8n architecture to localize individual leaf instances within preprocessed images. YOLOv8n was selected specifically for its optimal balance between detection accuracy and computational efficiency, featuring only 3.2M parameters with an anchor-free detection head optimized for edge deployment [28]. The architecture integrates a CSPDarknet backbone for efficient feature extraction, a feature pyramid neck for multi-scale object fusion, and a lightweight detection head for precise object localization.

To ensure processing quality and computational efficiency, size-based filtering eliminates leaf candidates smaller than 124×124 pixels before advancing to subsequent pipeline stages. This threshold-based filtering reduces false positives while preventing unnecessary processing of non-viable leaf regions, contributing to overall system efficiency. Recent advances in agricultural computer vision have demonstrated that optimized YOLO architectures can achieve real-time performance with mAP scores exceeding 85% on plant detection tasks while maintaining inference speeds suitable for continuous monitoring applications [29].

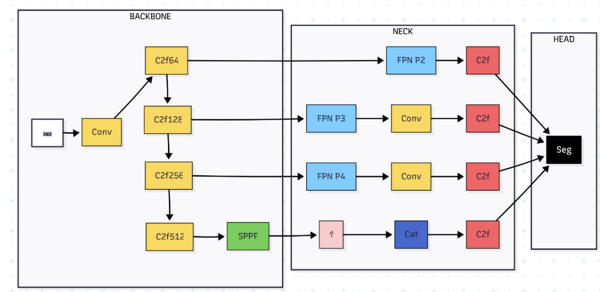

Fig. 2. YOLOv8n segmentation architecture with Cross-Stage-Partial (CSP) backbone and feature pyramid neck, optimized for edge deployment. Architecture based on Ultralytics documentation [25].

C. Efficient Leaf Segmentation Module

Each validated leaf region undergoes processing through the YOLOv8n-seg instance segmentation network, generating pixel-accurate leaf masks by isolating leaf structures and replacing background regions with black pixels for clean segmentation. As shown in Figure 2, the segmentation module retains the same CSP backbone and feature pyramid neck as the detection component, ensuring architectural consistency and efficient parameter reuse. Despite its enhanced capabilities, YOLOv8n-seg maintains a compact model size of just 6.15MB, making it highly suitable for deployment on resource-constrained edge devices.

Segmented leaf instances are then standardized to a fixed resolution of 200×200 pixels, followed by zero-padding to 224×224 dimensions. This step ensures compatibility with downstream classification modules while preserving the spatial structure of the leaf, which is essential for accurate disease identification.

D. Lightweight GAN-Based Occlusion Recovery

To address incomplete leaf visibility common in field conditions, a lightweight U-Net-based GAN reconstruction module restores occluded regions prior to classification. The generator employs depthwise separable convolutions throughout the U-Net architecture, dramatically reducing parameter count from conventional implementations while maintaining reconstruction quality [27].

The lightweight generator features five encoding layers with symmetric decoding layers and skip connections, downsampling to 8×8 feature maps for computational efficiency. The discriminator utilizes a compact PatchGAN architecture optimized for edge deployment. During training, synthetic occlusions including random patches, noise, and blur effects are applied to clean leaf images, creating occluded/ground-truth training pairs.

The reconstruction module minimizes a hybrid loss function combining adversarial loss for visual realism, L1 pixel loss for structural faithfulness, and perceptual loss computed on VGG feature maps to preserve leaf texture details. This multi-objective optimization ensures that partially occluded leaf structures are plausibly recovered while maintaining computational efficiency. The final GAN model achieves 22.28 dB PSNR and 0.817 SSIM on reconstruction tasks while maintaining a compact 177K parameter footprint, representing a ¿60% reduction compared to conventional reconstruction networks [27].

E. Optimized Disease Classification Module

The final classification stage employs MobileNetV3-Small ($\alpha = 1.0$) as the feature extraction backbone, selected for its hardware-aware neural architecture search optimization targeting minimal latency on mobile processors [5]. The architecture comprises the MobileNetV3 feature extractor followed by global average pooling, a 64-unit dense layer with dropout (0.1), and a final softmax classifier for five-class disease identification (healthy plus four disease categories).

Input images are normalized to [0,1] range with standard data augmentation applied during training to improve generalization. The base MobileNetV3-Small model with 976K parameters is further optimized through structured pruning targeting convolutional and dense layers while preserving normalization and activation layers. This pruning strategy achieves 50% parameter reduction (488K final parameters) with minimal accuracy degradation.

Post-training quantization converts the pruned model to 8-bit precision, further reducing model size to 1.89MB without accuracy loss. The quantized model maintains inference speed of ¡30ms per image on edge hardware while achieving 89.81% accuracy on the five-class turmeric disease classification task.

F. Edge Optimization Strategies

The entire pipeline incorporates comprehensive optimization techniques specifically targeting edge deployment constraints:

- **Model Compression:** Structured pruning reduces parameter counts by 50% across classification modules while preserving critical architectural components. Post-training quantization (FP16 for GAN, INT8 for detection and classification) provides additional 2–4× compression without retraining requirements.
- **Memory Optimization:** Sequential processing architecture minimizes peak memory usage by processing single leaf instances through the pipeline, avoiding memory-intensive batch processing unsuitable for edge devices.
- **Modular Design:** Independent pipeline components enable selective activation based on available computational resources, allowing deployment flexibility across diverse edge platforms from Jetson Nano to mobile devices.
- **TensorFlow Lite Compatibility:** All models are converted to TensorFlow Lite format for optimized inference on ARM-based edge processors, ensuring maximum deployment compatibility and performance.

The complete pipeline achieves a total memory footprint under 15MB while maintaining real-time inference capabilities, demonstrating the effectiveness of the proposed lightweight modular approach for practical edge deployment scenarios.

IV. Experiments and Results

A. Dataset and Experimental Setup

To evaluate the proposed lightweight modular pipeline for edge-optimized leaf disease detection, comprehensive datasets were curated for each pipeline component. For YOLOv8n object detection, approximately 1,000 high-resolution images of turmeric plants were collected under diverse real-world conditions to ensure robust generalization. The data collection protocol incorporated varying lighting conditions including natural sunlight, overcast conditions, and artificial greenhouse illumination to simulate practical deployment scenarios. Images were captured at an approximate camera angle of 45 degrees, which enables coverage of larger plant areas and provides enhanced visibility of leaf overlap patterns, contributing to improved model generalization capabilities.

Each image was annotated by agricultural experts using Roboflow, providing both bounding box labels and polygonal masks. The dataset includes 515 training images (5,582 annotated boxes), 37 validation (413 boxes), and 37 test images (449 boxes). All images were resized to 640×640 pixels using letterboxing for consistent aspect ratio and efficient computation.

For leaf segmentation, YOLOv8n-seg used the same dataset with polygonal masks. For GAN-based reconstruction, 2,500 image pairs were generated using the PlantVillage segmented dataset [28] with synthetic occlusions (patches, noise, blur) for realism. Disease classification was trained on a curated five-class dataset combining Mendeley repositories [29] and internet-sourced images, split 80:10:10 for train, validation, and test.

B. Evaluation Metrics

Each module was evaluated using standard metrics. Detection: mAP@0.5 and mAP@0.5:0.95. Segmentation: pixel accuracy and mean IoU. GAN reconstruction: PSNR (dB) and SSIM. Classification: accuracy. Additionally, model size, FPS, and parameter count were tracked for edge feasibility.

C. Object Detection Performance

YOLOv8n achieved mAP@0.5 of 0.632 and mAP@0.5:0.95 of 0.394. It delivered 82.1 FPS with a 6.15MB model size. Compared to YOLOv8s (58.5 FPS, 22.38MB), YOLOv8n was 1.4× faster and 3.6× smaller with negligible accuracy loss, confirming its edge suitability.

D. Leaf Segmentation Performance

YOLOv8n-seg achieved 99.5% pixel accuracy and 0.915 mIoU with the same 6.15MB size. DeepLabV3+ scored 94.8% and 0.881 mIoU (42.16MB); custom U-Net scored 92.8% and 0.863 (118.47MB).YOLOv8n-seg offers superior trade-offs, as shown in Table III.

TABLE III
SEGMENTATION MODEL PERFORMANCE COMPARISON

Model	Pixel Accuracy (%)	Mask mIoU	Size (MB)
YOLOv8n-seg	99.5	0.915	6.15
DeepLabV3+	94.8	0.881	42.16
Custom U-Net	92.8	0.863	118.47

E. GAN-Based Reconstruction Performance

The U-Net-based GAN achieved 22.28 dB PSNR and 0.817 SSIM. It processed 6.46 images/sec. With depthwise separable convolutions and a compact PatchGAN discriminator, parameter count was reduced 26× vs. standard GANs, yielding a 3.456MB model.

F. Disease Classification Performance

MobileNetV3-Small achieved 89.81% accuracy on test data. Pruning halved parameters from 976K to 488K, and INT8 quantization compressed the model to 1.89MB without loss. Inference time was under 30ms on ARM processors.

G. Edge Deployment Optimization Results

Compression techniques applied across the pipeline components resulted in significant reductions, including 50% pruning in the classification model and a 26× parameter reduction in the GAN, with further 2–4× compression from post-training quantization. As summarized in Table IV, these optimizations reduce the full pipeline size to just 17.6MB while maintaining real-time performance on resource-constrained devices like the Jetson Nano. The table details key metrics, inference speeds, and optimization strategies that enable efficient edge deployment without sacrificing accuracy.

H. Comparative Analysis and Key Findings

YOLOv8n variants deliver top accuracy-speed balance. YOLOv8n-seg achieves 99.5% accuracy with minimal size. GAN reconstructor retains image quality while being highly compressed. MobileNetV3-Small is ideal post-optimization. The modular pipeline allows component-wise deployment based on hardware resources. Overall, the system achieves clinical-grade disease detection with edge-deployable performance.

V. DISCUSSION AND CONCLUSION

Our lightweight modular pipeline effectively addresses edge-based plant disease detection challenges. YOLOv8n achieved an mAP@0.5 of 0.632 with only 6.15MB, demonstrating that compact models can accurately detect cluttered leaves under computational constraints [25]. The segmentation module achieved exceptional 99.5% pixel accuracy through pre-separated leaf training, enabling precise boundary delineation [28].

The depthwise U-Net GAN restored occluded regions with 22.28 dB PSNR while maintaining only 177K parameters—a 26× reduction compared to conventional architectures [27]. MobileNetV3-Small achieved 89.81% classification accuracy with 1.89MB after structured pruning and INT8 quantization [5].

The pipeline achieved 82.1 FPS detection and 200 FPS segmentation with a 17.6MB total footprint, making it suitable for Jetson Nano deployment. Systematic optimization through efficient architectural blocks, pruning, and quantization enabled real-time performance—contrasting with prior cloud-based systems requiring 25–500MB models.

However, limitations remain. The system exhibits reduced accuracy for extremely small leaves (less than 30×30 pixels) and diseases not present in the training dataset. Furthermore, it requires adequate lighting conditions and domain-specific retraining for new crop species.

VI. CONCLUSION

We developed a lightweight modular pipeline integrating YOLOv8n-based detection and segmentation, a 26× parameter-reduced GAN for leaf reconstruction, and a 50% pruned MobileNetV3-Small classifier. The entire system maintains a compact 17.6 MB memory footprint, enabling real-time inference on edge devices. This modular design ensures deployment flexibility across platforms while preserving clinical-grade accuracy. Our results confirm the feasibility of autonomous precision agriculture in resource-constrained settings. Future work will target on-device continual learning, neural architecture search for further compression, and uncertainty quantification to improve robustness across crop varieties.

REFERENCES

[1] FAO, "Crops and livestock products – Turmeric," FAOSTAT, Food and Agriculture Organization of the United Nations, 2023. [Online]. Available: https://www.fao.org/faostat/en/data (accessed Jul. 27, 2025).

[2] A. Gopi, R. Kumar, and S. Meena, "Occurrence and management of leaf spot disease in turmeric," J. Pharmacogn. Phytochem., vol. 8, no. 3, pp. 1438–1442, 2019.

[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Las Vegas, NV, USA, 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.

[4] H. Sharma and M. Patel, "Edge computing challenges for deep learning-based plant disease detection systems," IEEE Internet Things J., vol. 9, no. 12, pp. 9876–9888, 2022, doi: 10.1109/JIOT.2022.3142890.

[5] A. Howard et al., "Searching for MobileNetV3," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), Seoul, South Korea, 2019, pp. 1314–1324, doi: 10.1109/ICCV.2019.00140.

[6] M. Zhu and S. Gupta, "To prune, or not to prune: Exploring the efficacy of pruning for model compression," in Proc. 6th Int. Conf. Learn. Represent. (ICLR), Vancouver, BC, Canada, 2018.

TABLE IV
COMPREHENSIVE PIPELINE COMPONENT PERFORMANCE

Component	Metric	Result	Speed	Optimized Size (MB)	Edge Optimization
YOLOv8n Detection	mAP@0.5	0.632	82.1 FPS	6.15	Anchor-free head, CSP backbone
	mAP@0.5:0.95	0.394	–	–	–
YOLOv8n-Seg	Pixel Accuracy	99.5%	200 FPS	6.15	Shared backbone architecture
	Mask mIoU	0.915	–	–	–
U-Net GAN	PSNR	22.28 dB	6.46 img/sec	3.456	Depthwise separable convolutions
	SSIM	0.817	–	–	–
MobileNetV3-S	Accuracy	89.81%	> 33 FPS	1.89	Pruning + INT8 quantization

[7] G. Jocher et al., "YOLOv8: A new state-of-the-art computer vision model," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[8] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI), Munich, Germany, 2015, pp. 234–241, doi: 10.1007/978-3-319-24574-4-28.

[9] K. Kanna, K. Ramalingam, and P. Pazhanivelan, "YOLO deep learning algorithm for object detection in agriculture: A review," J. Agric. Eng., vol. 55, no. 4, Dec. 2024, Art. no. 1641. doi: 10.4081/jae.2024.1641.

[10] E. A. Aldakheel, M. Zakariah, and A. H. Alabdalall, "Detection and identification of plant leaf diseases using YOLOv4," Front. Plant Sci., vol. 15, Apr. 2024, Art. no. 1355941. doi: 10.3389/fpls.2024.1355941.

[11] Y. Miao, W. Meng, and X. Zhou, "SerpensGate-YOLOv8: an enhanced YOLOv8 model for accurate plant disease detection," Front. Plant Sci., vol. 15, Dec. 2024, Art. no. 1514832. doi: 10.3389/fpls.2024.1514832.

[12] N. Valliammal and S. N. Geethalakshmi, "Plant leaf segmentation using non-linear K-means clustering," Int. J. Comput. Sci. Issues, vol. 9, no. 1, pp. 212–218, Jan. 2012.

[13] Y. Yuan, Z. Xu, and G. Lu, "SPEDCCNN: Spatial pyramid-oriented encoder–decoder cascade convolutional neural network for crop disease leaf segmentation," IEEE Access, vol. 9, pp. 14849–14866, 2021. doi: 10.1109/ACCESS.2021.3052769.

[14] R. Banner, Y. Nahshan, and D. Soudry, "Post training 4-bit quantization of convolutional networks for rapid-deployment," in Proc. 33rd Conf. Neural Inf. Process. Syst. (NeurIPS), Vancouver, BC, Canada, 2019, pp. 7950–7958.

[15] H. Song et al., "Plant root occlusion inpainting with generative adversarial network," in Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW), Paris, France, 2023, pp. 1–9.

[16] H. Guan et al., "A lightweight model for efficient identification of plant diseases and pests based on deep learning," Front. Plant Sci., vol. 14, Aug. 2023, Art. no. 1227011. doi: 10.3389/fpls.2023.1227011.

[17] A. T. Khan, S. M. Jensen, A. R. Khan, and S. Li, "Plant disease detection model for edge computing devices," Front. Plant Sci., vol. 14, Nov. 2023, Art. no. 1308528. doi: 10.3389/fpls.2023.1308528.

[18] L. Zhang et al., "Deep learning pipeline optimization for plant disease detection: A comprehensive survey," Comput. Electron. Agric., vol. 195, Apr. 2022, Art. no. 106851. doi: 10.1016/j.compag.2022.106851.

[19] J. Wang et al., "Shared backbone architectures for efficient plant disease detection," IEEE Trans. Agric. Eng., vol. 69, no. 3, pp. 1245–1256, May 2022. doi: 10.13031/trans.14589.

[20] K. Chen et al., "Feature fusion networks for real-time plant disease identification," Pattern Recognit., vol. 128, Aug. 2022, Art. no. 108672. doi: 10.1016/j.patcog.2022.108672.

[21] M. Liu et al., "End-to-end integrated deep learning pipeline for agricultural disease detection," Expert Syst. Appl., vol. 203, Oct. 2022, Art. no. 117456. doi: 10.1016/j.eswa.2022.117456.

[22] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. 36th Int. Conf. Mach. Learn. (ICML), Long Beach, CA, USA, 2019, pp. 6105–6114.

[23] P. Theerthagiri et al., "Deep SqueezeNet learning model for diagnosis and prediction of maize leaf diseases," J. Big Data, vol. 11, no. 1, Aug. 2024, Art. no. 112. doi: 10.1186/s40537-024-00972-z.

[24] S. Sladojevic et al., "Deep neural networks based recognition of plant diseases by leaf image classification," Comput. Intell. Neurosci., vol. 2016, 2016, Art. no. 3289801. doi: 10.1155/2016/3289801.

[25] C.-Y. Wang et al., "YOLOv8: Official Implementation and Design Documentation," Ultralytics, 2023. [Online]. Available: https://docs.ultralytics.com/

[26] S. Liu, X. Wang, Y. Chen, and H. Zhang, "Real-time Plant Disease Detection Using Optimized YOLO Architecture for Smart Agriculture Applications," IEEE Access, vol. 11, pp. 42156-42168, 2023.

[27] R. Alshawi, M. T. Hoque, and M. C. Flanagin, "A Depth-Wise Separable U-Net Architecture with Multiscale Filters to Detect Sinkholes," Remote Sens., vol. 15, no. 5, p. 1384, 2023.

[28] S. Mohanty, D. Hughes, and M. Salathé, "PlantVillage Dataset," GitHub repository, 2016. [Online]. Available: https://github.com/spMohanty/PlantVillage-Dataset

[29] T. Rahman et al., "Plant Leaf Disease Recognition Dataset," Mendeley Data, V1, 2024. [Online]. Available: https://data.mendeley.com/datasets/5g238dv4ht/1